MILITARY SPECIFICATION

MOTOR GENERATOR, AC,FOR BAROSWITCH TESTER, T4016

1. SCOPE

1.1 This specification covers requirements, quality assurance provisions, and the preparation for delivery criteria for one type of alternating current motor generator entitled as Motor Generator AC for Baroswitch Tester, T4016 which is used with the Nike Hercules warhead section.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-E-5272	- Environmental Testing, Aeronautical and Associated Equipment, General Specification for.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-810	- Environmental Test Methods.
MIL-STD-1167	- Ammunition Data Card.
MIL-STD-1169	- Packaging, Packing and Marking for Shipment of Inert Ammunition Components.

DRAWINGS

ORDNANCE CORPS

8803988	- Motor Generator.
EL-8816847	- Equipment List.
8853954	- Test Panel.
8853955	- Operating Instructions.

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.2 Assembly. The motor generator shall comply with all requirements specified on the drawing (dwg) 8803988, with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the drawing and herein.

3.3 First article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

3.4 Serialization. The contractor shall assign serial numbers as required by drawings. The contractor may select any number which complies with the drawing requirement. These numbers shall not be duplicated.

3.5 Regulation. The motor generator shall be capable of maintaining an output voltage and output frequency within the tolerances, while under the conditions listed below, when tested as specified in 4.4.1.

Input Voltage	Ohms Load	Output Voltage	Output Frequency
28 ± 1%	22.6 ± 1%	26 ± 2.0 volts	400 ± 20
28 ± 1%	13.0 ± 1%	26 ± 3.0 volts	400 ± 20
26 ± 1%	22.6 ± 1%	26 ± 2.0 volts	400 ± 20
26 ± 1%	13.0 ± 1%	26 ± 3.0 volts	400 ± 20
30 ± 1%	22.6 ± 1%	26 ± 2.0 volts	400 ± 20
30 ± 1%	13.0 ± 1%	26 ± 3.0 volts	400 ± 20
30 ± 1%	no-load	31 volts max.	400 ± 20

3.6 Wave form. The total harmonic distortion, (defined as the square root of the sum of the squares of all voltage component harmonics) shall not exceed a maximum of five percent of the fundamental when tested as specified 4.4.2.

3.7 Dielectric strength. The motor generator, while at normal operating temperature, shall withstand without insulation breakdown, a 500-volt dc potential minimum applied for one minute between the dc input and the frame and between the dc and ac winding. It shall also withstand a 1000-volt dc potential minimum applied between ac output and frame when tested as specified in 4.4.3.

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3.8 Environmental testing. The motor generator shall comply with MIL-STD-810, visual inspection and failure criteria, section 3. The motor generator shall also comply with applicable paragraphs in this section before and after each environmental test. Testing shall be as specified in 4.4.5. Tests shall be conducted in the order specified.

3.8.1 High temperature. The motor generator shall show no evidence of deterioration, such as cracking, swelling, or other damage when subjected to a temperature of 165 + 5 degrees Fahrenheit °F when tested as specified in 4.4.4.1. After this test the motor generator shall conform to the requirements of 3.5 and 3.7.

3.8.2 Low temperature. The motor generator shall show no evidence of swelling, flaking, cracking, embrittlement or other damage when subjected to a temperature of -65 + 5°F when tested as specified in 4.4.4.2. After this test the motor generator shall conform to the requirements of 3.5 and 3.7.

3.8.3 Pull test. The leads shall withstand a six-pound pull test when attached to the motor generator when tested as specified in 4.4.4.3. After the test the motor generator shall conform to the requirements of 3.5.

3.8.4 Transportation vibration. The motor generator shall show no evidence of cracking, distortion, loosening or other damage when subjected to vibration in any direction at a frequency range of 55 to 10 Hz at a double amplitude of 0.060 inch when tested as specified in 4.4.4.4. After the test the motor generator shall conform to the requirements of 3.5 and 3.7.

3.8.5 Mechanical shock. The motor generator shall show no evidence of cracking, loosening, distortion or other damage after being subjected to an impact shock of 30 g when tested as specified in 4.4.4.5. After the test the motor generator shall conform to the requirements of 3.5.

3.8.6 Life cycle. The motor generator shall be capable of continuous operation for 500 hours under the conditions specified in 3.5. During this test the motor generator shall conform to the requirements of 3.6. The motor generator shall be capable of meeting four life cycles, with only brush changes permissible after each cycle, when tested as specified in 4.4.4.6.

3.8.7 Humidity. The motor generator shall show no evidence of obliterated marking, or deterioration such as swelling, corrosion, erosion, pitting, loosening of insulation or other damage when subjected to humidity conditions up to 95 percent at 165°F and when tested as specified in 4.4.4.7. After the test the motor generator shall conform to the requirements of 3.5 and 3.7.

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3.9 Workmanship.

3.9.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.9.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.9.3 Burrs. Parts shall be free from burrs.

3.9.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.9.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

3.9.6 Electrical. Cable assemblies and wires shall be dressed in such a manner as to prevent pinching or chafing of the wire insulation. Ends of wires shall be free from burrs, sharp edges, or points. All parts shall be free from flux, chips, dirt, grease, rust or foreign material.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection.

4.2.1 Quantity to be submitted. When the contract requires first article inspection, the contractor shall submit 3 complete assemblies (see dwg 8803988) for first article testing to the testing facility designated in the invitation for bids or request for proposal.

4.2.2 Inspection and testing to be performed. The first article inspection sample shall be subjected to the inspections and tests specified in Table I. Tests shall be conducted in order specified in Table I.

Table I
First article inspection and tests

Inspection or Test	Dest	Non-Dest
1. Classification of characteristics (see 4.3.2.1)		X

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2.	Regulation (see 3.5)		X
3.	Wave form (see 3.6)		X
4.	Dielectric strength (see 3.7)		X
5.	High temperature (see 3.8.1)		X
6.	Low temperature (see 3.8.2)		X
7.	Pull test (see 3.8.3)		X
8.	Transportation vibration (see 3.8.4)	X	
9.	Mechanical shock (see 3.8.5)	X	
10.	Life cycle (see 3.8.6)		X
11.	Humidity (see 3.8.7)	X	

4.2.3 Inspection equipment. The inspection and testing specified in 4.2.2 shall be accomplished by using the gages and test equipment specified in 4.3.4.1.

4.2.4 Rejection. If any item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

4.3 Quality conformance inspection.

4.3.1 Inspection lot formation. All items in each inspection lot shall be from one production line, one shift, one batch of products, one manufacturer of raw material, one manufacturer of components, one production method, and one manufacturing operation. The entire lot shall be submitted as a unit for inspection.

4.3.2 Product inspection examination. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table (see Table II and Table III), and inspected for all characteristics in the applicable classification. Any characteristics in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

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Table II
Major Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

Table III
Minor Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-9	all	14-16	12	28-33	16	77-121	20
10	9	17-18	13	34-41	17	122-273	21
11	10	19-22	14	42-54	18	over 273	22
12-13	11	23-27	15	55-76	19		

4.3.2.1 Classification of characteristics (see 3.2). Critical, major and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Motor generator (see dwg 8803988).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Mounting hole out of location	Gage
	102. Total length, max	Gage
	103. Outside diameter, max	Gage
	104. Length of lead, min	Gage
	105. Insulation of lead split or broken	Visual
	106. Marking missing misleading or unidentifiable	Visual
	107. Any component loose or missing	Visual/Manual
	108. Objective evidence relating to material missing or improper (see 3.1)	Visual

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Minor:

201. Protective finish damaged to the extent of exposing base metal.....Visual
202. Evidence of poor workmanship (see 3.9).....Visual

4.3.2.1.2 Preparation for delivery (see 5).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Preservation, packing, and packaging not in accordance with MIL-STD-1169	Visual
	102. Marking not in accordance with MIL-STD-1169	Visual
Minor:		
	201. Evidence of poor workmanship (see 3.9)	Visual

4.3.3 Testing. All tests shall be performed. Tests shall be performed in the order specified below:

	Test	Test Classification
a.	Regulation	Major
b.	Wave form	Major
c.	Dielectric strength	Major
d.	High temperature	Major
e.	Low temperature	Major
f.	Pull test	Major
g.	Transportation vibration	Major

Of these tests, g is considered destructive and samples so tested shall not be returned to the lot. Destructive samples are over and above the deliverable quantity of items required by the contract.

4.3.3.1 Regulation, wave form and dielectric strength. Each motor generator in the lot shall be tested in accordance with 4.4.1, 4.4.2 and 4.4.3. If any item fails to comply with 3.5, 3.6 and 3.7, the item shall be rejected and removed from the lot.

4.3.3.2 High temperature, low temperature, pull test. and transportation vibration. Two sample motor generators shall be selected at random from each lot. The sample shall be tested in accordance with 4.4.4.1, through 4.4.4.4. If any item fails to comply with 3.8.1 through 3.8.4, the lot shall be rejected.

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4.3.4 Acceptance inspection equipment.

4.3.4.1 Critical and major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions and calibration instructions designed or specified by the contractor. The contractor shall obtain approval of such designs, operating instructions and calibration instructions prior to use on the contract from Commander, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SARPA-QA-N-I as specified on DI-R-1714 (see 6.1.2).

4.3.4.2 Minor characteristics and tests. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

4.4.1 Regulation. The output voltages and frequencies shall be recorded for each of the tests.

4.4.1.1 Rated load. Apply an input voltage of 28 volts dc ± 1 percent to the motor generator and measure the output voltage and frequency across a 22.6 ohm ± 1 percent resistive load. The output voltage shall be 26 ± 2.0 volts ac rms and the output frequency shall be 400 ± 20 Hz. The output shall remain stable with no sustained voltage or frequency oscillation. Repeat the test except that the resistive load shall be 13.0 ohm ± 1 percent and the output shall be 26 ± 3.0 volts ac rms.

4.4.1.2 Minimum input. Perform test as in 4.4.1.1 using an input voltage of 26 volts dc ± 1 percent.

4.4.1.3 Maximum input. Perform test as in 4.4.1.1 using an input voltage of 30 volts dc ± 1 percent.

4.4.1.4 No load. Perform test as in 4.4.1.3 using no loads. The output voltage shall not exceed 31 volts ac rms. The above tests shall be performed utilizing test panel dwg 8853954 and Operating Instructions dwg 8853955.

4.4.2 Wave form. The output voltage wave form shall be observed with an oscilliscope or equivalent acceptable testing device while the motor generator is operating under conditions of 4.4.1. The total harmonic distortion (defined as the square root of the sum of the squares of all voltage component harmonics) shall not exceed five percent maximum of the fundamental.

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4.4.3 Dielectric strength. The motor generator while at normal operating temperature shall be tested with a potential of 500 volts dc minimum which shall be applied for one minute between the dc input and the frame and then between the dc and ac windings. Then a 1000-volt dc potential minimum shall be applied between the ac output and the frame. The voltage source shall be adequately interlocked to protect human life.

4.4.4 Environmental testing (see 3.8).

4.4.4.1 High temperature. The motor generator shall be placed in a suitable test chamber. The temperature within the test chamber shall be raised and stabilized (see 6.4) at 165 ± 5°F for 2 hours. The motor generator shall then be returned to ambient room temperature by removal from the test chamber. No forced cooling shall be employed in returning the motor generator to ambient room temperature. After completion of this test the motor generator shall be tested as specified in 4.4.1 and 4.4.3.

4.4.4.2 Low temperature. The motor generator shall be tested as specified in 4.4.4.1 except that the test chamber temperature shall be lowered to -65 ± 5°F. No forced warming shall be employed in returning the motor generator to ambient room temperature. After completion of this test the motor generator shall be tested as specified in 4.4.1 and 4.4.3.

4.4.4.3 Pull test. A force shall be applied at a point one inch from the free end of each lead. This force shall be applied in any direction, including the weakest, and shall be increased gradually to six pounds and held at this value for 30 seconds minimum. There shall be no evidence of damage to the leads or solder connection. After completion of the test the motor generator shall be tested as specified in 4.4.1.

4.4.4.4 Transportation vibration. The motor generator shall be secured to a vibration table and vibrated for one hour minimum in each of three mutually perpendicular planes through a frequency range of 10 to 55 Hz at a double amplitude of 0.060 ± 0.006 inch. The frequency shall vary uniformly from 10 to 55 to 10 Hz making approximately one cycle per minute. After the test the motor generator shall be visually examined for damage and tested as specified in 4.4.1.

4.4.4.5 Mechanical shock. The motor generator shall be mounted in such a manner as to simulate service mounting using a device such as specified in MIL-E-5272, Procedure II Shock test, and shall be subjected to 18 impact shocks minimum of 30 g each. Of this total, three shock impulses shall be applied in each direction along each of three mutually perpendicular axes of the motor generator. Each shock impluse shall reach its maximum amplitude within 5-1/2 milliseconds maximum and shall have a duration of 11 ± 1 milliseconds. After the test the motor generator shall be visually examined for damage and tested as specified in 4.4.1.

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4.4.4.6 Life cycle. The motor generator shall be mounted by normal means and operated under conditions of 4.4.1.1 continuously for 500 hours minimum and at the completion of this test shall be tested as specified in 4.4.1. The motor generator shall be capable of four such life cycles permitting only brush changes after each cycle.

4.4.4.7 Humidity. The motor generator shall be placed in a humidity test chamber and within a period of one hour the chamber temperature shall be raised to 165 ± 5°F and stabilized at that temperature for two hours. A relative humidity of 95 percent shall be maintained while the dry-bulb temperature is 165 ± 5°F. At the end of the two-hour period, the chamber temperature shall be reduced uniformly over a one-hour period, to between 68° and 95°F. The relative humidity during the period of decreasing temperature shall be 100 percent. The assembly shall be subjected to a total of ten such humidity cycles (one hour of increasing temperature at 95 percent humidity, two hours of temperature stabilization at 95 percent humidity, and one hour of decreasing temperature at 100 percent humidity). The water used in controlling the humidity shall be distilled with a pH value between 6.5 and 7.5 at 77°F. At completion of the tenth cycle, the assembly shall be removed from the test chamber and excessive moisture removed with dry, filtered compressed air. The motor generator shall then be tested as specified in 4.4.1 and 4.4.3.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. The motor generator shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169(MU).

5.1.2 Level C. The motor generator shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169(MU).

5.2 Packing.

5.2.1 Levels A and C. The motor generator shall be packed in accordance with the applicable instructions contained in MIL-STD-1169(MU).

5.3 Marking. Marking shall be in accordance with MIL-STD-1169(MU).

5.4 Data cards. Data cards shall be prepared in accordance with MIL-STD-1167.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements.

a. Title, number, and date of this specification.

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b. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.1.2 Contract data requirements (see 4.3.4.1 and 5.4). Data specified in 4.3.4.1 and 5.4 will be delivered as identified on a numbered DD Form 1664 when required on a DD Form 1423 incorported in the contract.

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment (see 4.3.4.1). The contractor shall obtain approval from the Commander, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SARPA-QA-N-I.

6.4 Temperature stabilization. The temperature is considered stabilized when the greatest peaks (both highest and lowest) continually observed over 5 minutes min. do not exceed the nominal temperature plus tolerance specified.

6.5 Supersession data. This specification supersedes the requirements of Purchase Description X-PA-PD-1037 dated 29 April 1959.

Custodian:
Army - PA

Preparing activity:
Army - PA

Project No. NUOR-A040

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